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Chapter 2

Data Modeling using ERD

Data Modeling نمذجة البيانات

* **Meaning** → first step and high level of database design.

* **Processing** → the capturing important concepts and rules, then representing them in a blueprint for designing.

* **Goals** → Describe

- Data contains
  ماهي البيانات
- Relationships
  العلاقة
  بين Entity
- Data Constraints (rules)
  القوانين

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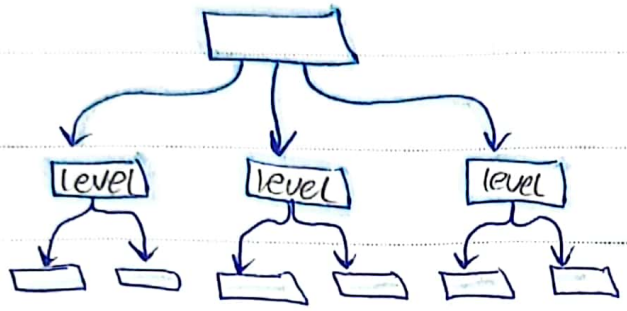
* Common database models

* Flat file model

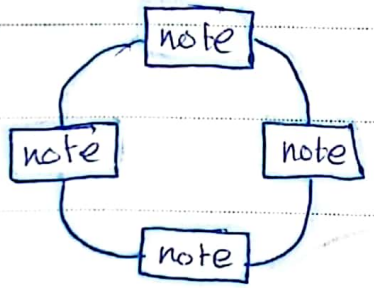
← مجموعة من الملفات

* Hierarchical model



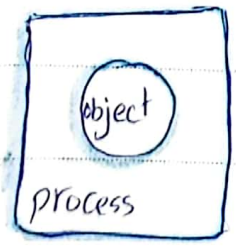
← like a tree

* Network model



لا تهتم لمستخدمها
إلا على
حسب الطلب

* object-oriented model



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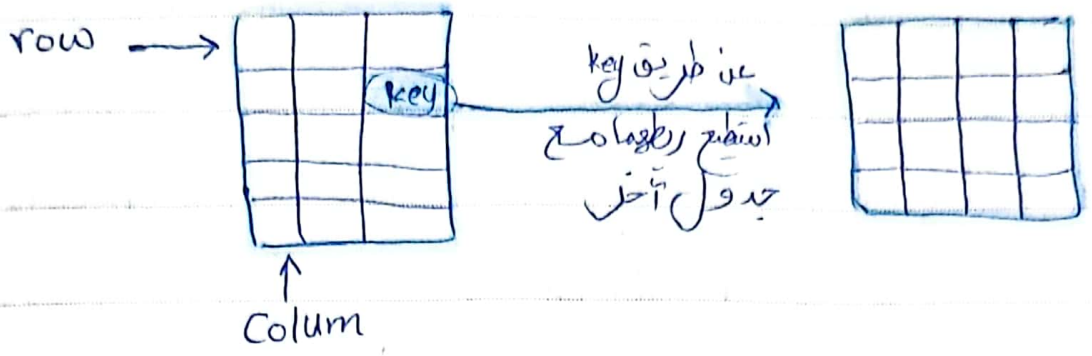
* Relational model



Data Development process DDP

- includes collecting and analyzing data for a business, (track and the diagramming) the organization in an Entity Relationship Diagram (ERD).

- Major components :

1. Conceptual Data Modeling
   نمذجة البيانات
   - Analyze.
   - Entity Relationship Diagram.

2. Database Design التصميم
   - Design.
   - Table, Definitions
     Index, View, Cluster.

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3. Database Build تحويلها لواقع
   ↓
   build

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Conceptual Model feature

* Captures business functional and informational needs.
* based (current needs), but reflect (future needs).
* addresses

does not addresses	addresses
implementation	need of a business
physically possible.	Conceptually ideal.

* includes all entities and relationships.
* ERM ⇒ entity relationship model.
* illustrated ERD.
* specifies attributes and relationship for E.

Entity Relationship Model (ERM)

- use to represents data requirement for a business and the type of database used.

- chen introduced the E-R Model in 1976.

* أميرة بأخلاقي

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- It includes all entities, attributes and importance relationships.

- ERM

Entities ⇓ tables holding specific information or data.

Relationship ⇓ defind associations or interactions between entities.

- detailed, logical representation of or organizational data shown in an ERD.

ERD (entity relationship diagram)

※ meaning:

is model that describ the concepts or entities that exist in a system and the relationship.

※ purposes:

1. serves as documentation tool.
2. communicate logical structure (database) to users.

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③ help analyst / desinger to understanding of database content through the construction.

④ effectively communicates logical database to users.

* Goals :

- required information.
- information appears only once.
- avoid modeling derivable information from another information.
- Locate information
  - predictable
  - logical place

* Build :

* Create entities and Attributes.
* Choose Unique Identifiers.
* Build Relationships.
* Identify optionalities and Cardinalities.
* check the model.

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عنصر Entities and Instance

- any item or anything like (person, place, thing or event or concept) that save inf
- in the user environment that the company wants to save the data about it.
- Represented as rectagles with entity names inside.

example

Person	Place	object	Event	Concept
employee	store	book	Sale	account
student	hospital	car	renewal	course
patient				

Instances → مثال عن العنصر

- is a single occurrence of an entity. هو الظهور الوحيد للـ entity

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Entities represent a set

→ instances that are of interest to a particular business.

Attributes

describe entities and are the specific information that must be known.

single-valued property of an entity.

represented by an oval.

Designing

entity

- rectangular shape with capitalized name inside.
- names must be differen except for foreign key /attribute.

attribute

oval shape with a capital first letter for the name.

→ Domain

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* No special characters (+, -, <, ?, !) only underscore allowed. (_).

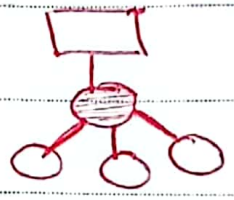
Attribute

simple / atomic

cannot be divided into subparts.

composite

divide subparts represent basic attributes.



Attribute

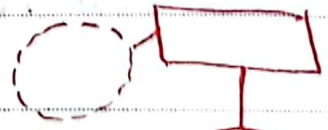
Multivalued

set of values for each entity (more than 1 v at one time.

Derived

values Calculated from the attributes,

stored attribute



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Valued attributes

Multi

can have more than 1 value at one time.

Single

can have a sigle value at a time.

Key Identifier

Values ~~used~~ used to uniquely indentify individual entity.

Relationship

bidirectional between 2 entities or between an entity and itself.

Type

set of associations entities of one or more entity type.

Instance

exactly one entity from each participating entity type.

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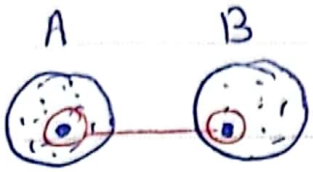
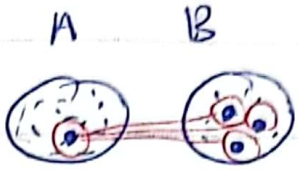
Relationship type

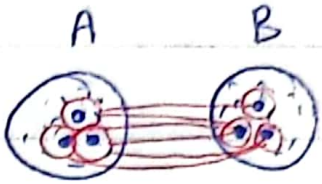
one to one



one to many



many to many



Degree of relationship type

number of entity

Unary

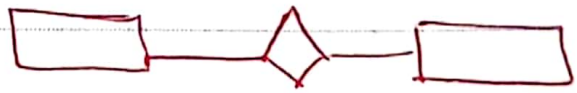
⇓

Contains one entity

Binary

⇓

two entities linked to each other



Cardinality Constraint

⇓

مدى ظهور Entity داخل العلاقة

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Cardinality Constraints

- many
- one
- zero

* measures the quantity of sth

قياس كمية شئ ما

* minimum Cardinality

⇓

zero or one

→ optional (zero)

→ mandatory (one) exactly.

* maximum Cardinality

⇓

one or many

Mandatory

one and only one

one or many

Optional

zero or one

zero or many

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Attribute one Relationship

⇓ show

many to many relationship

example :

- Task_Assigned.
- start_Date.

don't have key attribute. ⇐ **Weak entity**

⇓

Types of entity depends of other entity type.

have key attribute ⇐ **strong entity**

⇓

do not depends of other entity.

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